



Factorize, Don't Truncate

Physics-aware, graph-free leakage-minimizing dataset splitting at 10-million-molecule scale

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github.com/Ray16/PALM

Motivation

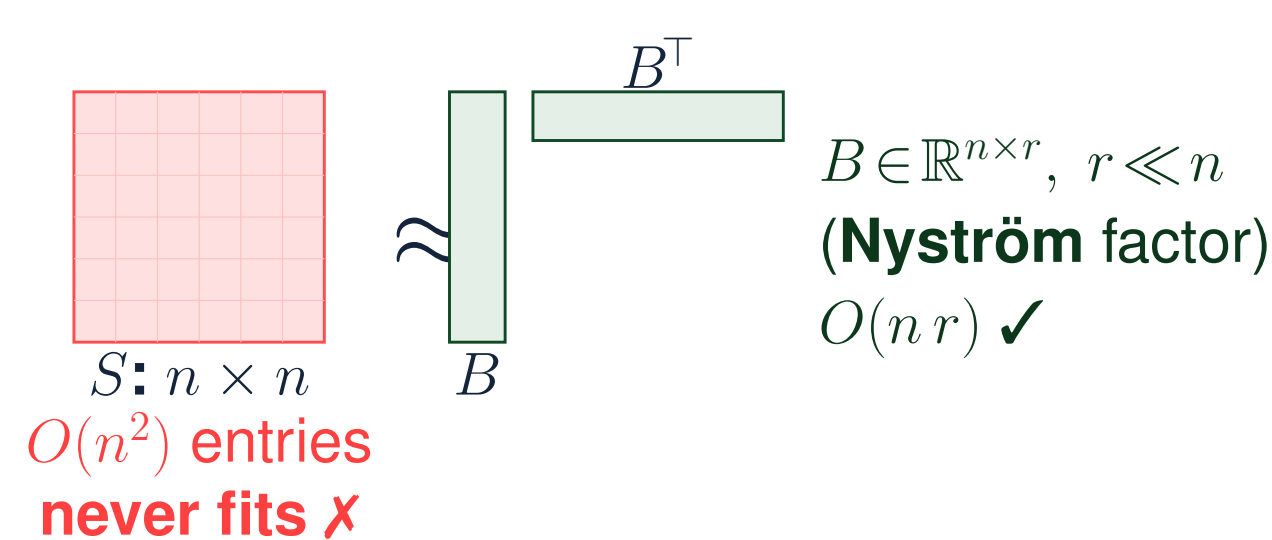
Data leakage inflates ML metrics. Split a dataset carelessly and near-duplicates land on both sides. The model then scores by *recall*, not by *generalization* — the benchmark rewards memorization, and the gap only surfaces on genuinely new chemistry.

Existing splitters cannot handle large datasets.

- **Exact methods** solve an $O(n^2)$ combinatorial problem — they stall orders of magnitude below today's datasets.
- **Fast methods** stay fast by **truncating** the similarity — optimizing a fraction of the signal and calling it a split.

So at 10^6 – 10^7 molecules — the scale that now matters — leakage goes **entirely unmeasured**.

Key idea



Tanimoto & cosine are **PD kernels** $\Rightarrow$ a Nyström factor B with $S \approx BB^T$ exists — and S is **never formed**: $O(nr)$ time and memory.

Leakage metric

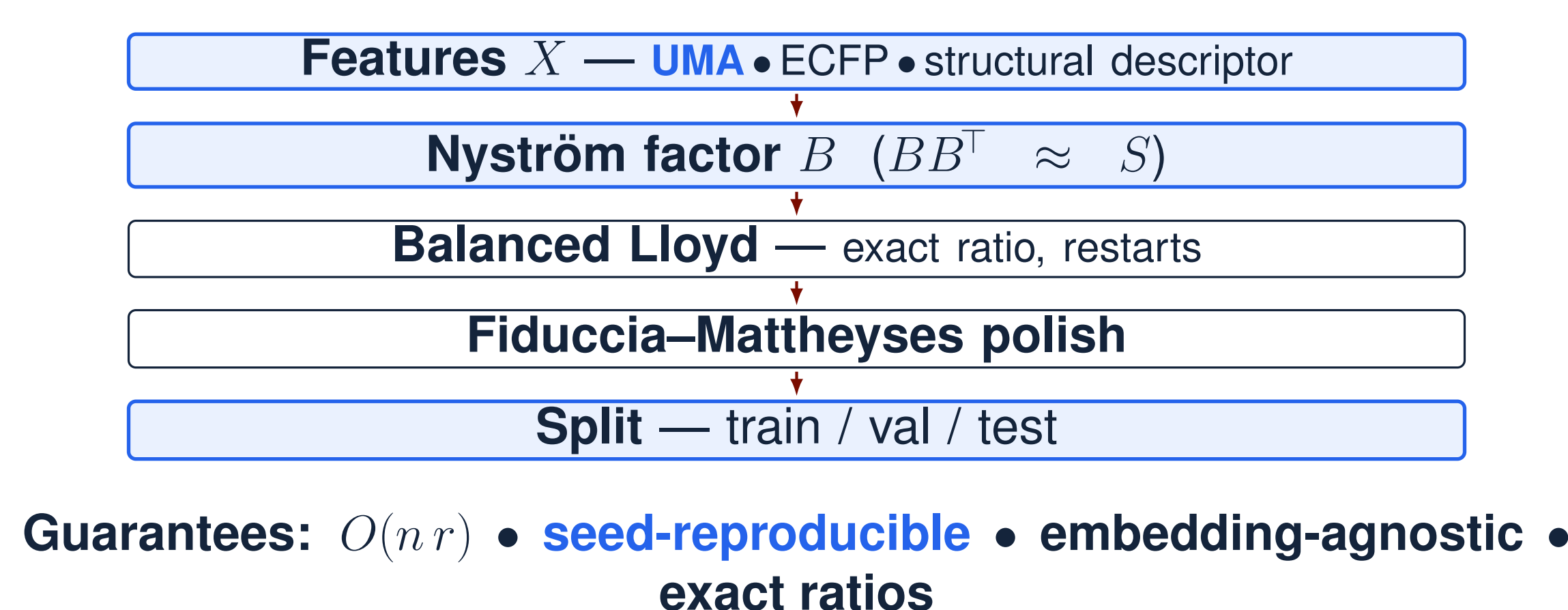
Fraction of total pairwise similarity crossing a split boundary. With **block sums** $p_c = \sum_{i \in c} B_i$ and $s = \sum_c p_c$:

$$L(\pi) = 1 - \frac{\sum_c \|p_c\|^2}{\|s\|^2}$$

Each p_c is an r -vector, so **any** candidate split is scored in $O(r)$ — this is what makes the optimizer's inner loop cheap.

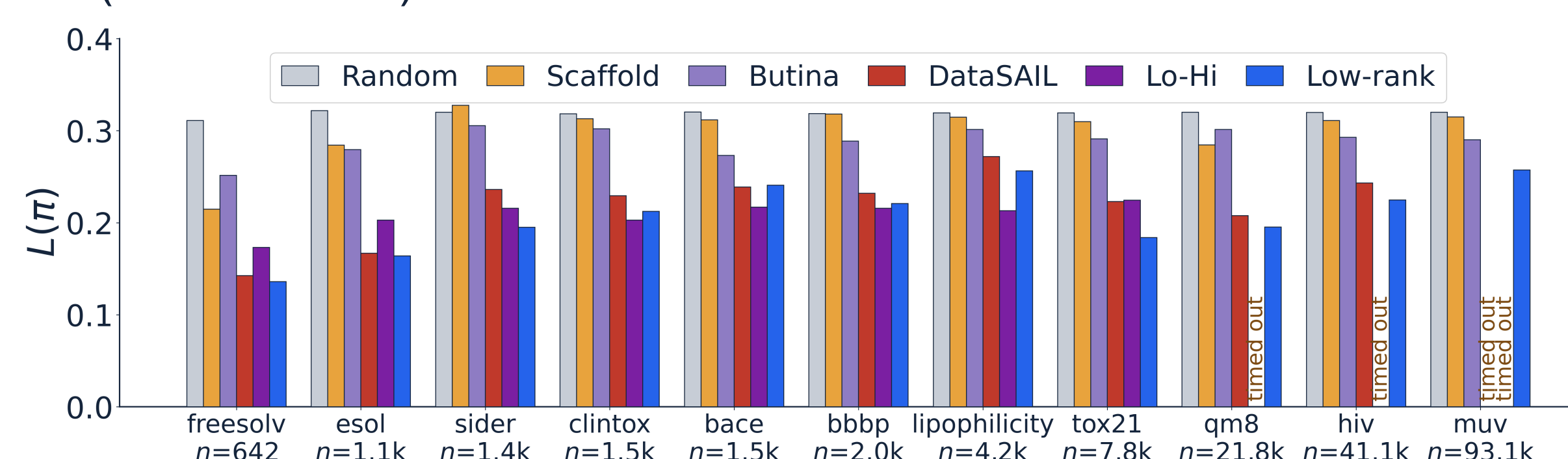
The factorization is faithful. Scoring a split through B instead of forming S changes the answer by $< 10^{-4}$: factorized $L(\pi) = 0.6573$ vs. exact $O(n^2) = 0.6573$.
validated on a 40k OMol25 cosine subsample at $r = 256$

Pipeline



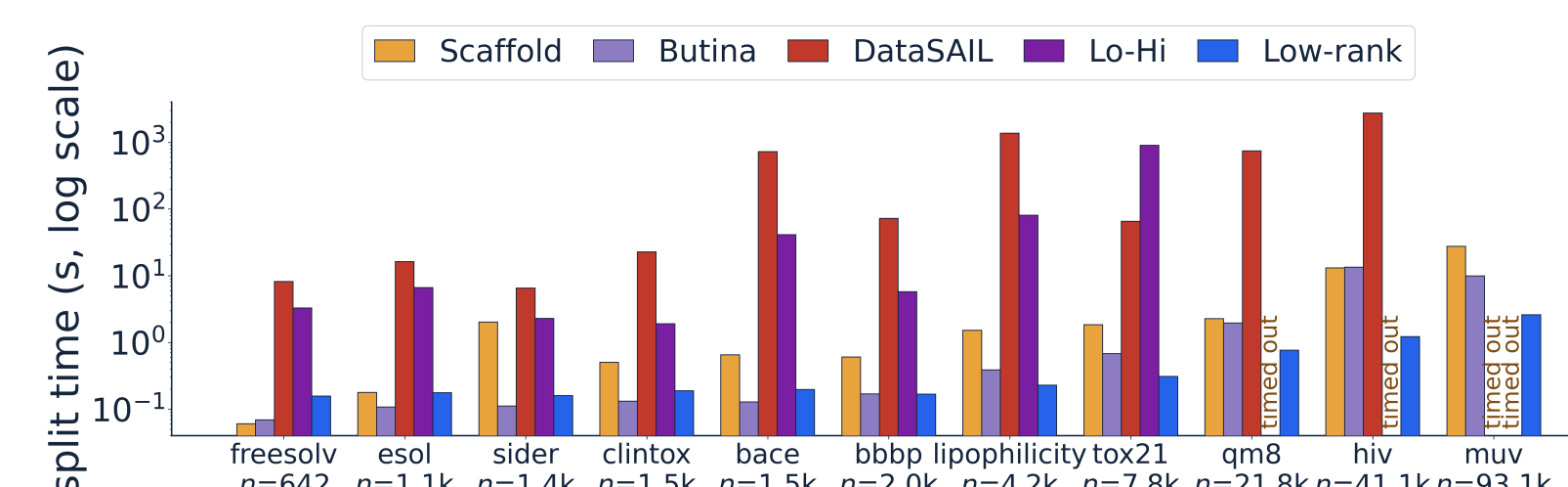
Preliminary results

(a) Lowest leakage of the fast splitters, 11/11. Each group is one MoleculeNet dataset, $L(\pi)$ on 1024-bit ECFP over 4 seeds (lower = cleaner). **Every** method runs at an **exact 80/20** block, so none can gain from a smaller test set. Low-rank (blue) is lowest of the fast splitters — random, scaffold, Butina — on **11/11**, and beats the exact methods DataSAIL & Lo-Hi on nearly all (table below).



Scaffold sits at the random baseline on 8/11 (*worse* than random on *sider*); **Butina** clustering helps only modestly. Only *optimizing the leakage objective* drops well below random.

(b) Fast where the exact methods are not. Split time, log–log. Low-rank splits 93k in **2.6 s**; ILP/MILP sit 1–3 orders above.



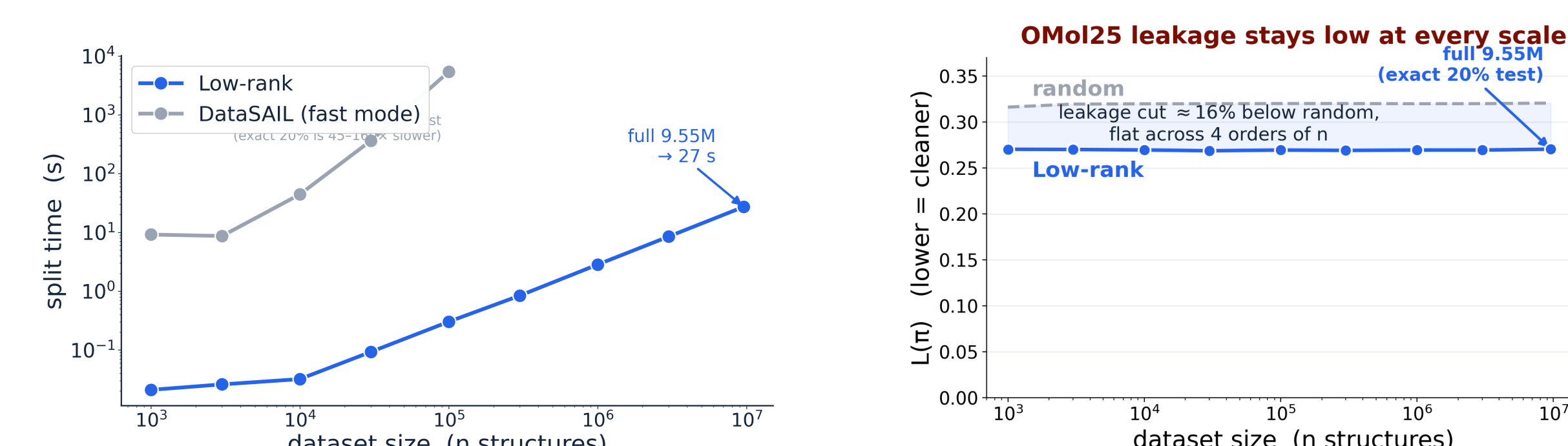
(c) Vs. the two exact leakage-aware methods.

	DataSAIL (ILP)	Lo-Hi (MILP)	Low-rank
Lower $L(\pi)$	1/10	0/8	9/10, 8/8
Split time	≤ 90 min	≤ 15 min	≤ 27 s
Largest n run	100k	7.8k	9.55M
Keeps all data	yes	no	yes

Scope: neither can be forced to exact 80/20, so $L(\pi)$ is matched on each method's own ratio — Lo-Hi on the subset it retains (it discards 0.1–2%). DataSAIL re-run locally; its only $L(\pi)$ win is *bace*. Time & largest- n include the OMol25 runs.

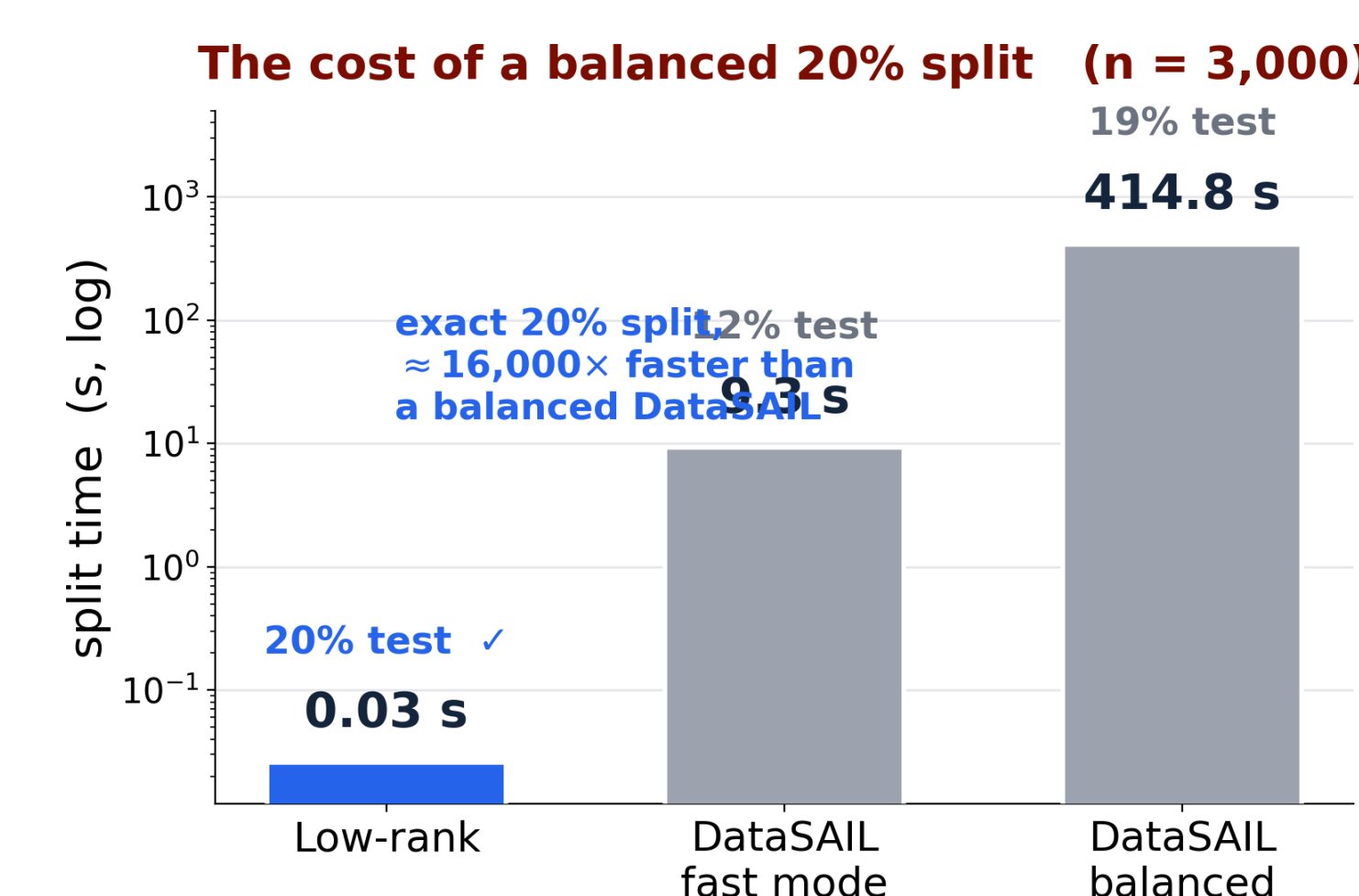
Lo-Hi: Steshin, S. Lo-Hi: Practical ML Drug Discovery Benchmark. NeurIPS 2023 Datasets & Benchmarks Track.

Scaling to 10^7 : time & leakage



Head-to-head with DataSAIL

Embedding-agnostic: it runs *directly* on OMol25's metal complexes and radicals. Scaffold, Butina & Lo-Hi need **SMILES** and *cannot run* here at all.



DataSAIL's **fast** mode collapses the test set to **5–12%** (never 20%); forcing $\approx 20\%$ needs fine clustering that is **45–160 $\times$** **slower** and *still* plateaus at 19%. Low-rank hits **exact 20%** in **0.03 s** and holds it to 9.55M.

Discussion

- **Factorize, don't truncate.** Scoring the *whole* similarity in $O(nr)$ (no k-NN truncation, no ILP) gives the **lowest $L(\pi)$ of the fast splitters** (11/11), deterministically, in seconds.
- **Scale, reach, exact balance.** First leakage-aware splitter *we know of* to reach 10^7 (9.55M in 27 s) and the **only one that runs on OMol25**; DataSAIL holds 80/20 only at 45–160 $\times$ the cost, while low-rank gives **exact 20%** at lower leakage.
- **Honest scope.** On an independent fingerprint the margins are *comparable*, not lower; downstream-error validation is future work.

27 s

split the full **9.55M**-molecule OMol25

2054 $\times$

faster than DataSAIL (GPU vs. CPU ILP)

11 / 11

lowest $L(\pi)$ of the fast splitters, exact 80/20

exact 80/20

every split, exact target ratios